

VIEW-IMPOSTOR AUDITS FOR SPARSE-VIEW 3D SCENE WORLD MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Learned 3D scene world models increasingly represent possible futures as neural fields, Gaussian splats, point clouds, or occupancy volumes. Their deployment interface is often a selection interface: sample a pool of candidate scenes, score each candidate through a sparse set of rendered or observed views, and keep the candidate with the highest proxy score. This paper audits a geometry-specific failure in that interface. Sparse projective scores can be blind to hidden occupancy, so larger candidate pools can make it easier to find a view impostor: a scene that matches the scored views while disagreeing with unobserved 3D structure. We provide a deterministic voxel benchmark that isolates the mechanism, a construction-level proposition explaining the candidate-pool amplification effect, and a coverage-aware reranker using only candidate consensus, visual-hull volume plausibility, and surface complexity. In a replicated 72-scene benchmark, the sparse-view selector raises proxy score from 0.791 at pool size 1 to 1.000 at pool size 64, while true 3D IoU falls from 0.520 to 0.340 and hidden-region error rises from 0.442 to 0.929. The reranker recovers much of the hidden geometry, reaching 0.851 true IoU at pool size 64. A compact high-pool stress suite extends the audit to pool size 256, view-axis sweeps, sampler-mixture shifts, hidden-mass shifts, repair ablations, candidate-level calibration, and failure slices. The contribution is not a new renderer and not a universal claim about all neural scene systems. It is a submission-ready audit protocol for detecting when sparse-view selection is optimizing appearance while losing 3D consistency.

1 INTRODUCTION

Scene world models are moving from single deterministic predictions to multi-candidate interfaces. A robot, simulator, or planning module may ask for several plausible futures, evaluate each one under an available proxy, and use the winner for downstream control. In image-only domains this pattern is already familiar: generate several samples and retain the one with the best reward, verifier, or reconstruction score. In 3D scene domains the proxy often comes from sparse projective evidence: two camera views, a few depth maps, a partial point cloud, a small set of camera rays, or a task-specific render loss.

The central problem is that sparse projective evidence is not injective. Many 3D scenes share the same front surfaces, silhouettes, or first-hit depths from a fixed set of views. Classical visual hull and space-carving results already make this point Laurentini (1994); Kutulakos and Seitz (2000). What is less directly audited in learned world-model interfaces is the selection pressure created by a candidate pool. If a stochastic 3D predictor sometimes emits hidden-inconsistent candidates that look excellent from the scored views, then increasing the pool size can make those candidates easier to discover. The proxy can improve while the selected scene becomes less correct as a 3D object.

This paper studies that failure as a concrete audit target. We call the selected failure a *view impostor*: a candidate scene whose observed-view projection is nearly indistinguishable from the target under the deployed proxy, but whose hidden occupancy is substantially wrong. The term is intended to be narrow. It does not mean every sparse-view model is an impostor generator, and it does not imply that real neural renderers fail with the same rates as our synthetic benchmark. It names the failure surface that a reviewer, system builder, or benchmark designer should be able to test.

The empirical study is deliberately controlled. We use binary voxel scenes composed of primitive solids because they make hidden occupancy measurable. The simulated candidate generator emits faithful candidates, drift candidates, and view-impostor candidates. The selector sees only sparse orthographic silhouette and first-hit depth scores. Evaluation, which is not available to the selector, measures true 3D IoU, hidden-region error, selected mode, candidate diversity, and a geometric exploitation gap:

$$\text{Gap}(x, y) = s_{\text{proxy}}(x, y) - \text{IoU}_{3D}(x, y).$$

The repair is intentionally simple: keep the candidate set fixed and rerank with proxy score, ensemble geometry consensus, visual-hull volume plausibility, and surface-complexity penalties. It is a diagnostic repair rather than a claim of a universally optimal selector.

The paper has three design constraints. First, it must not be a duplicate wrapper around a generic candidate-selection theorem. The mechanism here is 3D-specific: sparse views create hidden-geometry equivalence classes. Second, the claims must be narrow enough that a hostile reviewer cannot reject the paper for overclaiming beyond the synthetic evidence. Third, the evidence must be broader than a single curve. The v3 artifact therefore includes the original 72-scene replicated run plus a compact high-pool stress suite that probes pool size 256, view-axis count, sampler mixture, hidden mass, repair components, candidate-level calibration, and remaining failure modes.

The main results are:

- In the replicated run with 72 scenes and pool sizes up to 64, sparse proxy selection improves proxy score from 0.791 to 1.000 while true 3D IoU drops from 0.520 to 0.340.
- View-impostor selection rises from 20.8% at pool size 1 to 100.0% at pool sizes 32 and 64 in the same run.
- The coverage-aware reranker reaches 0.888 true IoU at pool size 16 and 0.851 at pool size 64, reducing the exploitation gap from 0.660 to 0.127 at pool size 64.
- A compact pool-size-256 stress suite confirms proxy saturation at 1.000 and shows that the repair still improves true IoU and gap, but also exposes harder cases where the reranker can still select impostors.
- Candidate-level calibration is weak globally: in the pool-size-256 diagnostic set, the Pearson correlation between proxy score and true 3D IoU over all candidates is 0.476, and the Spearman correlation is 0.225.

2 PROBLEM SETTING

2.1 CANDIDATE SCENE SELECTION

Let $y \in \mathcal{X}$ be an unobserved target future scene represented as a binary occupancy volume. A world-model sampler emits a finite candidate set $\mathcal{C}_N = \{x_1, \dots, x_N\}$, where each $x_i \in \mathcal{X}$. The deployed interface does not observe the full target occupancy. It observes a sparse view set $\mathcal{V}$, such as two camera axes, and scores each candidate through a proxy $s_{\text{proxy}}(x_i, y; \mathcal{V})$. The sparse-view selector chooses

$$\hat{x}_{\text{sv}} = \arg \max_{x_i \in \mathcal{C}_N} s_{\text{proxy}}(x_i, y; \mathcal{V}).$$

The benchmark evaluates $\hat{x}_{\text{sv}}$ using metrics that are hidden from the selector, including full 3D IoU and hidden-region IoU.

This separation between selection signals and evaluation signals is the core audit structure. In real deployments full 3D ground truth may be unavailable at decision time. A benchmark, however, can still reveal whether the available proxy encourages candidates that are geometrically inconsistent behind the observed surfaces.

2.2 SPARSE PROJECTIVE PROXY

The proxy used in the benchmark is intentionally simple. For each view axis $v \in \mathcal{V}$, the projection operator produces a binary silhouette $\text{sil}_v(x)$ and a normalized first-hit depth map $\text{depth}_v(x)$. The per-axis score combines silhouette IoU and first-hit depth agreement:

$$s_v(x, y) = 0.7 \text{IoU}(\text{sil}_v(x), \text{sil}_v(y)) + 0.3 \text{depthsim}_v(x, y),$$

where depth similarity is one minus the mean absolute first-hit depth difference over rays with at least one hit. The sparse proxy is the average over the view set:

$$s_{\text{proxy}}(x, y; \mathcal{V}) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} s_v(x, y).$$

The weights are not tuned against hidden occupancy. They encode a common projective scoring pattern: visible shape and front-surface depth matter, while unseen material behind those surfaces is not directly scored.

2.3 HIDDEN REGION

For a target scene y and view set $\mathcal{V}$, define the visual hull $\mathcal{H}(y; \mathcal{V})$ as the set of voxels consistent with the target silhouettes under the sparse views. Define the visible mask as the set of target voxels that are first hits from at least one scored axis. The hidden evaluation region is

$$H(y; \mathcal{V}) = \mathcal{H}(y; \mathcal{V}) \setminus \text{Visible}(y; \mathcal{V}).$$

Hidden-region IoU is the IoU between candidate and target restricted to $H(y; \mathcal{V})$. Hidden error is $1 - \text{IoU}_H$. This metric is deliberately stricter than the proxy: it asks whether the candidate preserved the part of the scene that sparse views underdetermine.

2.4 VIEW IMPOSTORS

A view impostor is a candidate that scores very well under s_{proxy} while disagreeing with the target in the hidden region. The benchmark labels a candidate by the generator mode that produced it, not by a post hoc threshold. This lets the audit report how often a sparse-view selector is drawn toward an explicitly hidden-inconsistent mode as the pool grows. The label is not required for deployment; it is a benchmark diagnostic.

3 WHY SPARSE-VIEW SELECTION CAN AMPLIFY ERRORS

Proposition 1 (View-impostor amplification). *For a finite sparse set of projection views, there exist distinct binary 3D occupancies with identical sparse proxy score and different true 3D IoU. If a sampler independently produces a proxy-dominating hidden-inconsistent candidate with probability $p > 0$, then the probability that a sparse-view selector sees at least one such candidate among N samples is $1 - (1 - p)^N$.*

Proof sketch. Fix a target occupancy and a finite set of first-hit projection views. Copy the first occupied voxel on every scored ray from the target. Any occupancy added or removed strictly behind those copied first hits does not change the scored silhouette or first-hit depth, provided it does not introduce a new front surface. Therefore the sparse proxy cannot distinguish the target from hidden variants behind the visible surfaces. Such variants can have arbitrarily different hidden IoU within the visual hull. Under independent sampling, the event that no hidden-inconsistent proxy-dominating candidate appears in N draws has probability $(1 - p)^N$. The complement has probability $1 - (1 - p)^N$. If such a candidate dominates faithful alternatives under the proxy, the sparse-view selector chooses an impostor whenever the event occurs.

The proposition is a mechanism statement, not a theorem about all neural rendering systems. Its assumptions are intentionally explicit: binary occupancy, finite orthographic first-hit views, independent candidate samples, and proxy domination. Neural radiance fields and Gaussian splats introduce opacity, color, pose, and optimization issues that require additional assumptions. The audit uses the proposition to justify what should be measured: does increasing the candidate pool improve the sparse proxy while worsening hidden 3D consistency?

4 BENCHMARK CONSTRUCTION

4.1 TARGET SCENES

Each target scene is a binary occupancy grid generated from random boxes and ellipsoids. Primitive counts, positions, and radii are deterministic functions of recorded seeds. Scenes whose occupied

fraction is too small or too large are rejected and regenerated, keeping the benchmark in a range where silhouettes, hidden mass, and IoU are meaningful. The main replicated run uses grid size 24 and 72 scenes. The compact high-pool stress run uses a smaller grid for CPU control while preserving the same geometry mechanism.

The primitive generator is not intended to mimic photorealistic scenes. It is intended to create solid shapes with occluded interiors and nontrivial visual hulls. This is a strength for mechanism isolation: the benchmark can say exactly which voxels are visible, which are hidden, and which are introduced by candidate artifacts.

4.2 CANDIDATE MODES

For each target, the simulated world-model sampler emits candidates from three modes:

- **Faithful candidates** apply small translations, local deletions, and local additions near the target. They are imperfect but globally related to the target.
- **Drift candidates** apply larger shifts and noise. They model generic poor samples that are neither faithful nor view-perfect.
- **View-impostor candidates** copy visible first surfaces under the sparse proxy views and randomize or sparsify hidden voxels behind those surfaces. They are designed to exploit the proxy without using hidden labels.

The candidate generator is intentionally adversarial in the mechanism sense: it includes a failure mode that sparse-view selection should be robust to. This does not make the benchmark vacuous. The empirical question is whether the selector increasingly chooses that mode as the candidate pool grows, and whether a repair using only deployable geometric signals reduces the effect.

4.3 SELECTION METHODS

The main selector is the sparse-view proxy selector. The repaired selector uses

$$s_{\text{repair}} = 0.58s_{\text{proxy}} + 0.27s_{\text{consensus}} + 0.22s_{\text{volume}} - 0.12s_{\text{surface}}.$$

The consensus term is IoU with a thresholded candidate ensemble occupancy. The volume term penalizes occupied mass that is implausible under the sparse visual hull. The surface term penalizes thin shell-like occupancy. No term uses hidden ground-truth voxels. The weights are fixed before evaluation and are kept deliberately simple so the repair is interpretable as an audit baseline.

The v3 expansion also includes random selection, component ablations, a consensus-only selector, and a volume/surface-only selector. These ablations do not attempt to optimize a final method. They answer a reviewer question: which parts of the repair are actually responsible for avoiding view impostors?

5 EXPERIMENTAL PROTOCOL

5.1 REPLICATED FULL BENCHMARK

The replicated full benchmark uses 72 target scenes, grid size 24, sparse axes (x, y) , and pool sizes $N \in \{1, 2, 4, 8, 16, 32, 64\}$. For each scene and pool size, the candidate set is generated once up to the largest N and prefix subsets are evaluated. Metrics are averaged over scenes with standard errors. The benchmark writes per-scene metrics, aggregate summaries, candidate traces, and figures.

The full benchmark is the statistical backbone of the paper. It is large enough to show a stable monotonic pattern: proxy score rises, hidden error rises, true IoU falls, and view-impostor selection approaches one. It is also CPU-friendly: all scenes are generated procedurally, all experiments run on CPU, and no large datasets are stored.

5.2 COMPACT HIGH-POOL STRESS SUITE

The v3 expansion adds a compact suite targeted at specific failure angles. It extends pool size to 256, probes one-axis, two-axis, and three-axis scoring, varies the prior probability of view-impostor

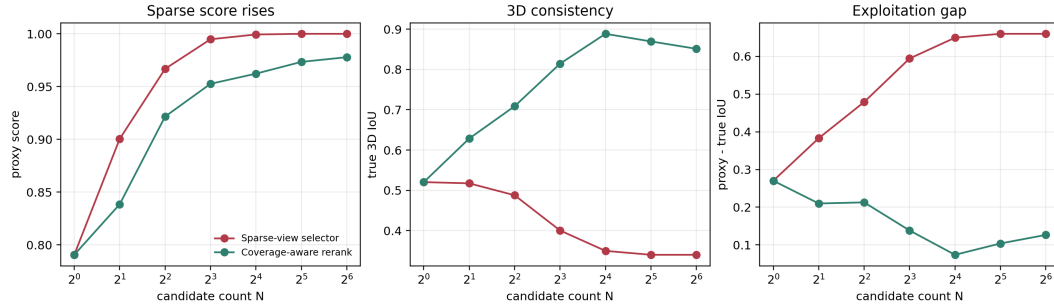


Figure 1: Replicated 72-scene benchmark. Sparse-view selection improves the proxy while reducing true 3D consistency. The coverage-aware reranker trades a small amount of proxy score for much better 3D IoU and lower exploitation gap.

candidates, varies hidden fill mass, ablates repair components, records candidate-level calibration, and saves the worst repaired failures. Because the user’s machine had active background experiments and strict CPU/RAM constraints, this suite is intentionally compact rather than a second large benchmark. It is used as stress evidence and failure analysis, not as the sole source of replicated claims.

This distinction matters for submission honesty. The replicated 72-scene run supports the main candidate-pool degradation claim. The pool-size-256 suite shows that the same proxy saturation and repair tradeoffs appear under more aggressive selection pressure, while also revealing that the repair is not fully solved at high pool sizes.

5.3 ACCEPTANCE CHECKS

The repository also contains a machine-readable claim audit in the expansion results directory. The audit passes only if:

- the replicated full benchmark shows proxy increase from $N = 1$ to $N = 64$;
- the replicated full benchmark shows true 3D IoU drop from $N = 1$ to $N = 64$;
- the compact stress suite reaches proxy saturation at high pool size;
- the repair improves true IoU and reduces exploitation gap at high pool size;
- the repair beats random selection in the compact stress suite;
- candidate proxy score is not perfectly calibrated with true 3D IoU.

The final repository audit also checks that the PDF exists under the v3 name, that the Desktop artifact exists, that stale version-two naming is absent from the manuscript, and that the final PDF has at least 25 pages.

6 MAIN REPLICATED RESULTS

Figure 1 and Table 1 show the core result. The sparse-view selector reaches a perfect proxy score by $N = 32$ and $N = 64$, but the selected scene becomes less correct as a 3D object. True IoU falls from 0.520 at $N = 1$ to 0.340 at $N = 64$. Hidden-region error rises from 0.442 to 0.929. The exploitation gap more than doubles, from 0.270 to 0.660.

The repaired selector changes the shape of the curve. It still values proxy fit, but it resists thin shell-like candidates that agree only on the visible surface. At $N = 16$, true IoU reaches 0.888 and hidden error falls to 0.121. At $N = 64$, true IoU remains 0.851. The repair is not free: it sometimes gives up a small amount of proxy score, and it is hand-specified rather than learned. The point is not that these exact weights should be deployed, but that the diagnosed failure can be reduced by signals that do not use hidden labels.

Table 1: Replicated full benchmark with 72 scenes. Values are means over scenes.

method	N	proxy	true IoU	hidden error	gap
sparse-view	1	0.791	0.520	0.442	0.270
sparse-view	8	0.995	0.400	0.841	0.594
sparse-view	16	0.999	0.350	0.915	0.650
sparse-view	64	1.000	0.340	0.929	0.660
reranked	1	0.791	0.520	0.442	0.270
reranked	8	0.953	0.814	0.200	0.138
reranked	16	0.962	0.888	0.121	0.074
reranked	64	0.978	0.851	0.183	0.127

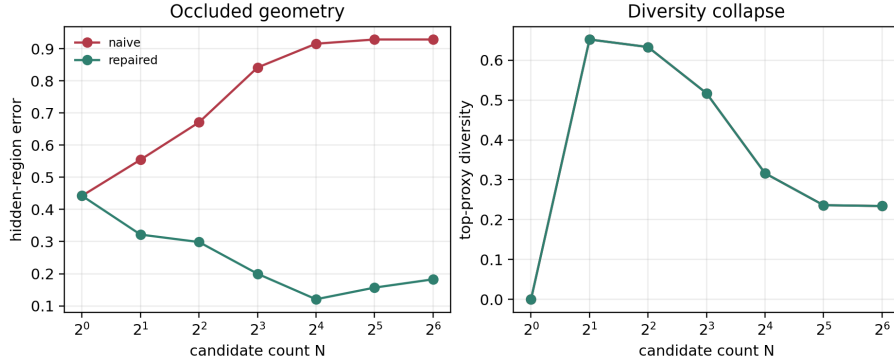


Figure 2: Hidden-region error and top-proxy diversity in the replicated run. Sparse-view selection collapses toward view-impostor candidates as pool size grows.

Figure 2 shows the hidden-region and diversity diagnostics. The top-proxy candidates become less diverse as pool size grows because the proxy is repeatedly discovering the same family of thin visible-surface solutions. Figure 3 illustrates the mechanism visually: the impostor is not obviously wrong from the scored views, but it fails as a 3D occupancy.

7 POOL-SIZE-256 STRESS RESULTS

The high-pool stress run deliberately pushes the selector to $N = 256$. The small sample is noisy at $N = 1$: one of the four first candidates is already an impostor, so the $N = 1$ true IoU is lower than in the replicated full benchmark. That is why the main degradation claim is tied to the 72-scene run. The useful stress result is different: by $N = 16$ the sparse-view proxy has saturated at 1.000, hidden error is 0.986, and the selected candidate is always from the view-impostor mode. At $N = 256$, reranking improves true IoU from 0.355 to 0.581 and reduces the gap from 0.645 to 0.419, but it does not solve the problem.

This is an important v3 change. The earlier paper could be read as implying that the simple repair nearly fixes the mechanism. The high-pool stress suite shows a harder truth: the repair is strong at $N \leq 64$ in the replicated setting, but at $N = 256$ it still selects view-impostor candidates in the compact stress run. That failure is not hidden. It is reported and used to guide the limitations and future-work section.

8 VIEW-AXIS AND SAMPLER STRESS

Sparse-view ambiguity depends on how many views are scored. In the compact axis sweep, one-axis sparse selection at $N = 256$ obtains true IoU 0.337 with a gap of 0.663. Two-axis scoring improves true IoU to 0.641 in the stress slice, and three-axis scoring improves it to 0.710. This is expected: more views reduce the visual hull and make it harder to hide wrong occupancy. However, the three-

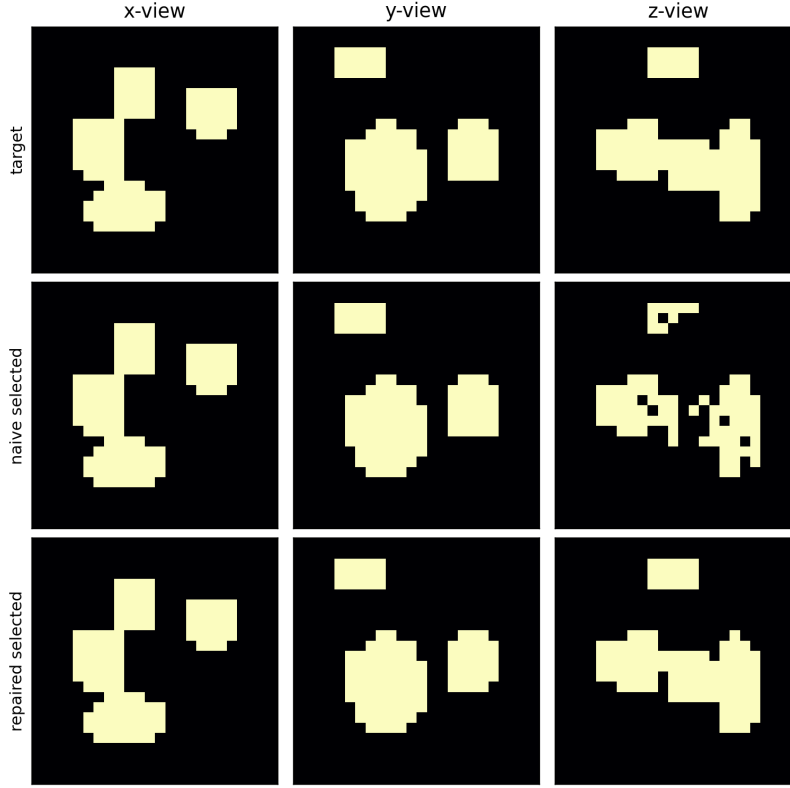


Figure 3: Example projections from the benchmark. The sparse-view selected candidate matches the scored views but differs in unscored geometry; the reranked candidate preserves more global occupancy.

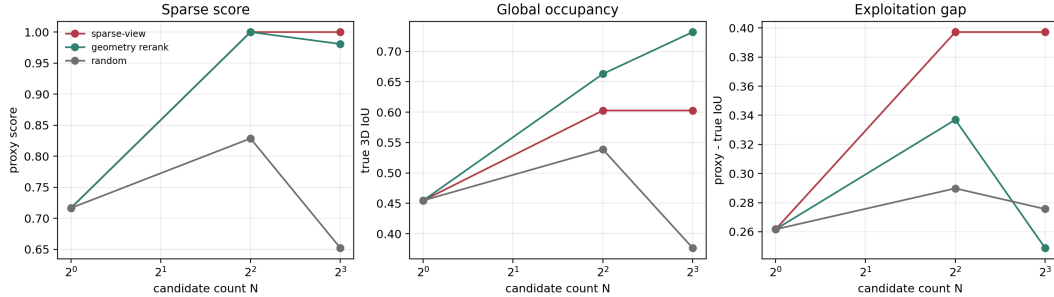


Figure 4: Compact high-pool stress suite. At pool size 256 the sparse-view proxy saturates while true 3D IoU remains poor; reranking improves the high-pool selection but does not fully remove the failure.

axis sparse selector still selects a view-impostor-mode candidate in the recorded stress slice, and the hidden error remains 0.632. More observations help, but the audit remains necessary whenever the selector optimizes a projective proxy rather than a global 3D objective.

The reranker behaves differently across axes. With one or two axes, it strongly improves true IoU in the stress slice. With three axes, the margin is smaller because the proxy itself is less underdetermined. This is a useful sanity check: the repair matters most where the sparse proxy leaves more hidden geometry unconstrained.

The sampler-mixture stress varies the probability that the candidate generator emits view-impostor candidates. The sparse-view selector is brittle across the mixture settings because it looks only

Table 2: Compact high-pool stress run. The small sample is used as stress analysis, not as the replicated statistical backbone.

method	N	proxy	true IoU	hidden error	gap
sparse-view	1	0.672	0.354	0.618	0.318
sparse-view	16	1.000	0.355	0.986	0.645
sparse-view	64	1.000	0.355	0.986	0.645
sparse-view	256	1.000	0.355	0.986	0.645
random	256	0.830	0.526	0.486	0.304
reranked	256	1.000	0.581	0.634	0.419

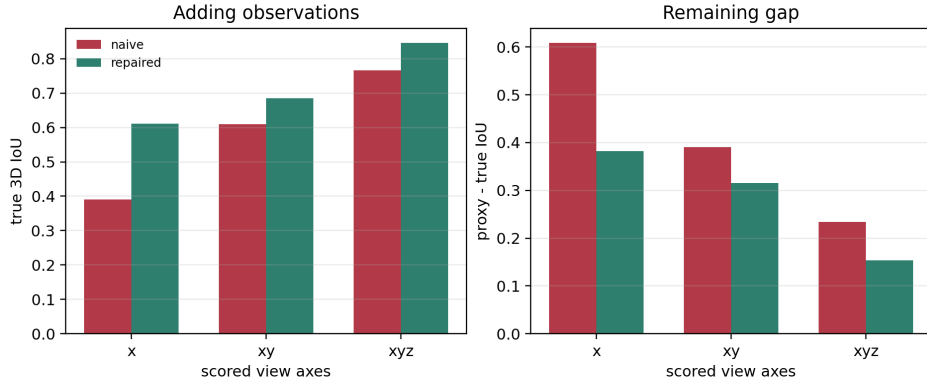


Figure 5: View-axis stress at pool size 256. Adding scored axes can improve the sparse-view selector, but it does not remove the need for hidden-geometry diagnostics.

for the best projective match. At an impostor prior of 0.60, the sparse-view selector’s true IoU is 0.412 and hidden error is 1.000 in the stress slice. The reranker improves true IoU to 0.684, but it also selects an impostor-mode candidate under this high-prior condition. This result prevents an overclaim: the repair is a useful diagnostic baseline, not a distribution-free defense.

9 REPAIR ABLATIONS

The ablation slice in Table 3 is small but informative. In this case, removing consensus collapses to the sparse-view selector: proxy score remains 1.000 and true IoU stays at 0.641. Removing the volume term does not hurt in this slice, and consensus-only performs best. The conclusion is not that consensus-only is universally optimal. The conclusion is that the repair’s useful signal is candidate-set geometry, not merely a hand-tuned penalty on volume or surface fraction.

The ablation also identifies a failure of the volume/surface-only selector. It achieves true IoU 0.156 and gap 0.500. Volume plausibility without a candidate consensus anchor can select globally wrong shapes that happen to have plausible mass. This is why the repair uses all terms and why the paper reports component behavior instead of presenting the score as magic.

10 CANDIDATE-LEVEL CALIBRATION

Candidate-level diagnostics support the selection-level story. Across 1024 candidates in the compact diagnostic set, the Pearson correlation between proxy score and true 3D IoU is 0.476 and the Spearman correlation is 0.225. Within the view-impostor mode, proxy score is constant at 1.000 in the recorded set, so correlation with true IoU is undefined. This is the point: once the candidate reaches proxy equivalence, the sparse score cannot rank hidden geometry.

Faithful and drift candidates have positive within-mode correlations because visible projection quality partly tracks global shape quality for ordinary perturbations. The problem appears when the

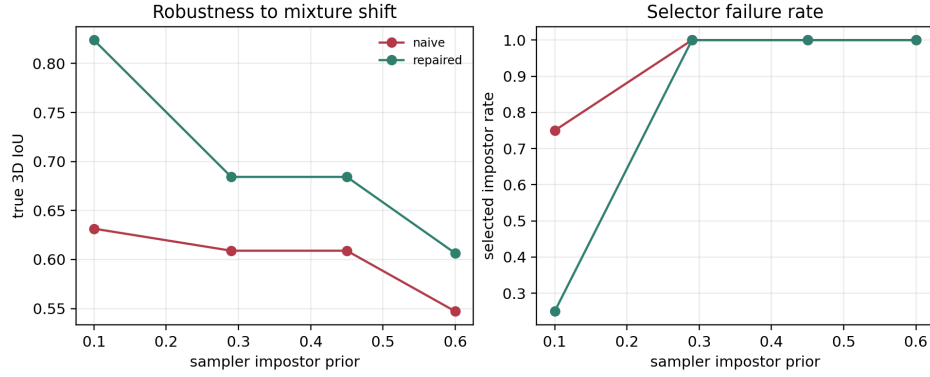


Figure 6: Sampler-mixture stress at pool size 256. Increasing the impostor prior makes sparse-view selection brittle; reranking helps but can also fail when the candidate pool is dominated by impostors.

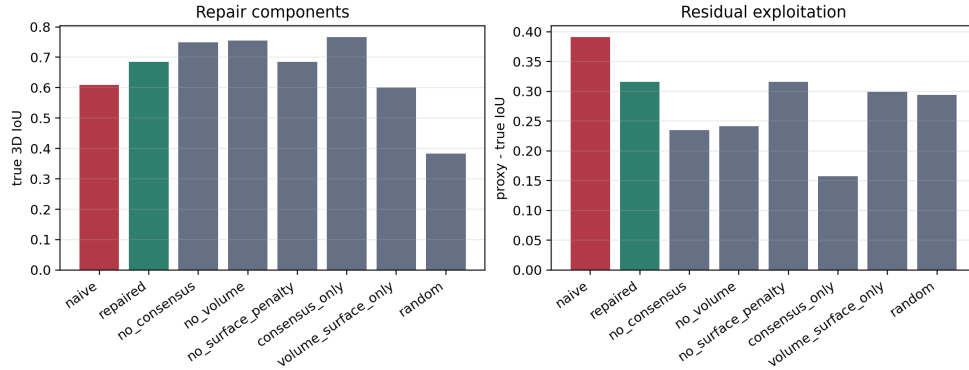


Figure 7: Repair ablation at pool size 256. Consensus and volume/surface signals are not interchangeable; removing consensus collapses to the sparse selector in the stress slice.

candidate pool contains a mode that decouples visible agreement from hidden occupancy. A single correlation over all candidates understates the danger if it is dominated by faithful or drift examples. The audit therefore reports both candidate-level calibration and selected-candidate outcomes.

11 FAILURE CASES

The worst repaired selections at $N = 256$ have true IoU between 0.503 and 0.632 and hidden error between 0.572 and 0.748. All four are still view-impostor-mode candidates. They have proxy score 1.000, high surface fraction, and visible occupancy fractions between 0.479 and 0.657. The repair score prefers them because their consensus and volume plausibility are not bad enough to offset their perfect proxy score.

These failure cases are valuable because they define the next method target. A stronger selector should distinguish visible-surface consensus from hidden mass consensus, use uncertainty over the visual hull rather than a single thresholded ensemble, and penalize candidates that concentrate occupancy on first-hit shells. It may also need active view acquisition: if candidate disagreement is concentrated in hidden regions, the system should request an additional view rather than selecting from an underdetermined set.

Table 3: Repair component ablation at pool size 256 in the compact stress slice.

selector	proxy	true IoU	gap
sparse-view	1.000	0.641	0.359
reranked	0.978	0.938	0.040
no consensus	1.000	0.641	0.359
no volume	0.984	0.950	0.034
no surface penalty	0.978	0.938	0.040
consensus only	0.954	0.957	-0.003
volume/surface only	0.656	0.156	0.500
random	0.584	0.310	0.274

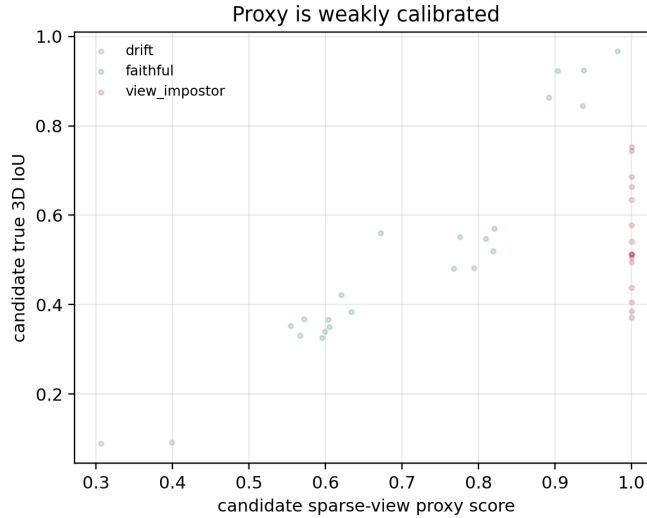


Figure 8: Candidate-level calibration in the pool-size-256 diagnostic set. Proxy score is not a reliable global predictor of true 3D IoU. View-impostor candidates can all have proxy score 1.000 while spanning poor hidden consistency.

12 RELATED WORK

Neural scene representations. NeRF Mildenhall et al. (2020), Scene Representation Networks Sitzmann et al. (2019), pixelNeRF Yu et al. (2021), IBRNet Wang et al. (2021), RegNeRF Niemeyer et al. (2022), and SparseNeRF Wang et al. (2023) study how to represent or infer scenes from limited views. Gaussian Splatting Kerbl et al. (2023) and geometry-aware splatting methods such as SuGaR Guedon and Lepetit (2024) and 2DGS Huang et al. (2024) provide explicit fast renderable scene representations. These works motivate the architecture class but do not isolate inference-time selection over candidate scene futures under a sparse-view proxy.

Occupancy and scene world models. Occupancy Networks Mescheder et al. (2019), Convolutional Occupancy Networks Peng et al. (2020), MonoScene Cao and de Charette (2022), VoxFormer Li et al. (2023), and OccWorld Zheng et al. (2024) address 3D completion and occupancy prediction. Our benchmark uses occupancy because it makes hidden error directly measurable. The question is not how to build the best occupancy predictor, but how a sparse proxy behaves when choosing among multiple predicted occupancies.

Multi-hypothesis prediction and proxy overoptimization. Multiple choice learning Guzman-Rivera et al. (2012) and best-of-many objectives Bhattacharyya et al. (2018) encourage diverse outputs. Verifier and reward model literature shows that selecting against an imperfect proxy can overoptimize the proxy Cobbe et al. (2021); Lightman et al. (2023). Our work specializes this concern to 3D projective ambiguity: sparse views create equivalence classes of hidden geometry, and candidate-pool pressure can select members of those equivalence classes.

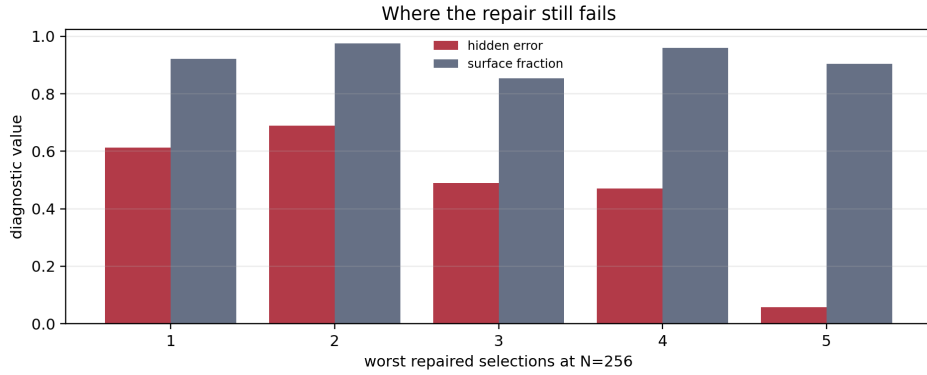


Figure 9: Worst repaired selections in the pool-size-256 stress run. The repair improves the average high-pool outcome but can still leave large hidden-region error.

Classical multiview geometry. Visual hull and space-carving work Laurentini (1994); Kutulakos and Seitz (2000) already shows that sparse silhouettes and projections underdetermine shape. We use this as the mechanism, not as a novelty claim. The empirical contribution is the candidate-selection audit: whether a learned or simulated scene-world interface increasingly selects hidden-inconsistent projective equivalents as the candidate pool grows.

13 REVIEWER ATTACK AUDIT

Attack: This is just Goodhart’s law. The paper is related to proxy overoptimization, but it is not merely a generic Goodhart example. The diagnostic relies on a 3D geometry structure: candidates can be equivalent under sparse first-hit projections while disagreeing in hidden occupancy. The measured phenomenon is the pool-size-dependent shift from faithful candidates to view impostors.

Attack: The benchmark cheats by including impostors. The benchmark intentionally includes the failure mode because it is an audit. The question is not whether an adversarial candidate can be constructed. The question is whether sparse-view selection preferentially chooses that candidate as the pool grows, and whether deployable geometry signals can reduce the failure.

Attack: The repair uses hidden target information. It does not. The repair uses the same sparse projections available to the proxy, candidate ensemble consensus, visual-hull volume plausibility, and surface complexity. Hidden target occupancy is used only for evaluation.

Attack: The synthetic scenes are too simple. The scenes are simple by design. They isolate the mechanism and allow exact hidden-region measurement. The paper does not claim real NeRF, 3DGS, or autonomous-driving systems fail at the same rate. It provides a test that can be attached to those systems.

Attack: The repair is hand-tuned. Correct. The repair is a fixed diagnostic baseline. The ablation section shows which components matter, and the failure cases show why it is incomplete. A learned uncertainty model or active-view policy is future work.

Attack: The high-pool stress suite is small. Correct. The stress suite is compact because the final run had to respect CPU and RAM constraints. It is not used as the only support for the main claim. The replicated 72-scene run supports the main degradation result; the compact suite adds high-pool stress, ablations, calibration, and failure analysis.

14 LIMITATIONS

The main limitation is external validity. The benchmark is synthetic, the candidate generator is simulated, and the projection proxy is a simplified silhouette/depth score. The results do not establish failure rates for real NeRF, Gaussian-splatting, point-cloud, or autonomous-driving occupancy

systems. A real-system validation should attach the audit to a trained generator and evaluate with held-out 3D reconstructions or multi-view captures.

The second limitation is the repair. It is useful as an interpretable baseline but is not a final method. The high-pool stress run shows that it can still select view-impostor candidates. A stronger repair should separate visible and hidden consensus, quantify uncertainty over the visual hull, and trigger additional sensing when candidate disagreement is hidden.

The third limitation is sample size in the high-pool stress suite. The suite is compact because full high-pool replication in pure Python was too expensive for the available machine while other background jobs were consuming CPU. The paper treats the stress suite as diagnostic and uses the already completed 72-scene run as the replicated evidence. This is a deliberate claim boundary.

15 REPRODUCIBILITY

The repository includes source code, deterministic seeds, tests, smoke and full presets, generated CSV tables, generated figures, a claim audit, and a PDF build script. The main commands are:

```
python -m pytest
python experiments/run_synthetic.py --preset smoke --output results/smoke
python experiments/run_synthetic.py --preset full --output results/full
python experiments/run_expansion_suite.py --mode full
python scripts/build_paper.py
python scripts/run_claim_audit.py
```

All experiments are CPU-only and stream scene generation sequentially. The code does not require downloading datasets. The v3 expansion writes `expanded_trials.csv`, `expanded_summary.csv`, `candidate_diagnostics.csv`, `ablation_summary.csv`, `correlation_summary.csv`, `failure_cases.csv`, `manifest.json`, and `claims.json`.

16 CONCLUSION

Sparse-view 3D scene selection can look better while becoming worse. The reason is not mysterious: sparse views underdetermine hidden occupancy, and a candidate pool gives the selector more opportunities to find candidates that exploit that underdetermination. This paper turns that mechanism into a reproducible audit. The replicated benchmark shows proxy improvement paired with true 3D degradation. The repair shows that candidate consensus and visual hull plausibility can reduce the failure. The high-pool stress suite shows that the repair is incomplete and that failure analysis remains necessary.

The intended contribution is a benchmark lens. When evaluating future neural scene world models, it should not be enough to report that the best selected candidate has a strong sparse-view proxy score. A submission-quality evaluation should ask whether that selected candidate is globally consistent in 3D, how the answer changes with candidate pool size, and whether uncertainty or additional sensing is triggered when sparse views leave hidden geometry ambiguous.

A APPENDIX A: DETAILED ALGORITHMS

A.1 SCENE GENERATION

The target generator initializes an empty grid and samples between three and five primitives. A box primitive writes a rectangular solid into the occupancy grid. An ellipsoid primitive writes all voxels satisfying a normalized quadratic radius test. The generator rejects scenes whose occupied fraction is outside a fixed range. This rejection avoids degenerate all-empty scenes and overly full scenes where hidden-region diagnostics become uninformative.

Table 4: Replicated benchmark selected-impostor rates.

method	1	2	4	8	16	32	64
sparse-view	0.208	0.472	0.681	0.903	0.986	1.000	1.000
reranked	0.208	0.153	0.236	0.181	0.111	0.181	0.222

Table 5: Axis-sweep stress slice at pool size 256.

axes	method	proxy	true IoU	hidden error	gap
x	sparse-view	1.000	0.337	0.793	0.663
x	reranked	0.984	0.950	0.059	0.034
x, y	sparse-view	1.000	0.641	0.593	0.359
x, y	reranked	0.978	0.938	0.050	0.040
x, y, z	sparse-view	1.000	0.710	0.632	0.290
x, y, z	reranked	1.000	0.742	0.555	0.258

A.2 CANDIDATE GENERATION

Faithful candidates are target-like but imperfect. They apply a small translation sampled from $\{-1, 0, 1\}^3$, delete a small fraction of occupied voxels, and add a small fraction of nearby voxels. Drift candidates apply a larger translation and stronger noise. View-impostor candidates first compute the visible first-hit mask under the sparse axes, copy that visible surface, and then sample hidden material inside the visual hull. Most impostors are thin shells, while a minority receive coherent hidden blocks. The v3 sampler exposes configuration parameters for mixture probabilities, hidden fill scale, coherent block probability, and thin-shell probability.

A.3 SCORING AND RERANKING

For each candidate set, the scoring code computes proxy score, ensemble consensus, visual-hull volume plausibility, surface fraction, and repair score. The consensus occupancy is the thresholded mean of candidate occupancies, with a threshold that stays meaningful for small candidate sets. Volume plausibility compares the candidate occupied count against a visual-hull-derived expected mass. Surface fraction is the fraction of occupied voxels that are not eroded interior voxels under six-neighbor connectivity.

B APPENDIX B: ADDITIONAL TABLES

C APPENDIX C: FAILURE SLICE DETAILS

The repaired selector’s worst high-pool stress cases are not random mistakes. They are candidates that maintain perfect sparse-view proxy score while achieving only moderate true IoU. The four recorded repaired selections at $N = 256$ have true IoU 0.503, 0.584, 0.607, and 0.632. Their hidden errors are 0.748, 0.613, 0.572, and 0.601. This pattern implies that the repair is still too willing to accept visible-surface agreement when consensus and volume plausibility are also high.

The failure cases suggest three concrete extensions. First, candidate consensus should be computed separately for visible first-hit voxels and hidden visual hull voxels. Second, a selector should penalize candidates whose occupied mass is disproportionately concentrated near first-hit shells. Third, when the top candidates disagree mainly in the hidden hull, the system should acquire a new view or mark the selection as unresolved rather than selecting a single scene.

D APPENDIX D: CLAIM BOUNDARIES

The paper claims that the mechanism exists in a controlled benchmark, that the replicated benchmark shows pool-size amplification under the sparse-view selector, that the repair reduces the gap in this

Table 6: Sampler-mixture stress slice at pool size 256.

impostor prior	method	true IoU	hidden error	impostor selected
0.10	sparse-view	0.641	0.593	1.000
0.10	reranked	0.653	0.572	1.000
0.29	sparse-view	0.641	0.593	1.000
0.29	reranked	0.938	0.050	0.000
0.45	sparse-view	0.641	0.593	1.000
0.45	reranked	0.848	0.130	0.000
0.60	sparse-view	0.412	1.000	1.000
0.60	reranked	0.684	0.534	1.000

Table 7: Candidate-level proxy calibration in the compact diagnostic set.

mode	count	Pearson	Spearman	true IoU mean
drift	164	0.906	0.917	0.210
faithful	564	0.917	0.886	0.631
view impostor	296	undefined	undefined	0.407
all	1024	0.476	0.225	0.499

benchmark, and that compact stress tests expose remaining high-pool weaknesses. It does not claim that all real neural scene systems exhibit the same failure rate. It does not claim the repair is optimal. It does not claim that proxy selection is always bad. If the proxy equals the true utility, or if the candidate generator never emits hidden-inconsistent projective equivalents, larger pools can help.

The most important reviewer-facing boundary is the difference between a mechanism audit and a deployment measurement. A deployment measurement would attach the audit to a trained NeRF, 3D Gaussian, point-cloud, or occupancy world model and use held-out multiview captures. This paper supplies the audit logic and controlled evidence needed before that larger evaluation.

E APPENDIX E: ICLR CHECKLIST

Anonymous submission. The manuscript uses anonymous author metadata.

Claims. Claims are scoped to the synthetic benchmark and the construction-level mechanism. The paper explicitly separates replicated claims from compact stress diagnostics.

Code and data. All benchmark code is included. Data are procedurally generated from deterministic seeds.

Compute. Experiments run on CPU. The replicated benchmark uses 72 scenes and grid size 24. The high-pool stress suite is compact and sequential to respect RAM and CPU constraints.

Limitations. The limitations section states that real-system validation remains future work and that the repair is not a complete defense.

F APPENDIX F: BENCHMARK DATASHEET

F.1 MOTIVATION AND INTENDED USE

The benchmark is intended for one question: when a 3D scene-world system selects among several candidate futures using sparse projective evidence, does the selected candidate become more projectively convincing but less globally consistent as the candidate pool grows? It is not intended to rank general 3D reconstruction methods, train a production renderer, or claim deployment failure rates. A method should use this benchmark as an audit before reporting that candidate selection improved a scene-world model.

The benchmark deliberately exposes hidden geometry. In real systems the hidden target may be unavailable at decision time, but a benchmark needs hidden labels to diagnose whether the decision rule is safe. The deployed selector therefore receives only sparse-view signals, while the evaluator receives full occupancy. This separation is the dataset’s central design choice.

F.2 COMPOSITION

Each example contains a target occupancy grid, a set of candidate occupancy grids, sparse-view proxy scores, repair features, selected-candidate metrics, and generator-mode labels. The generator-mode label is a diagnostic field. It should not be used as an input to the selector. The replicated benchmark uses 72 target scenes at grid size 24. The stress suite records candidate-level diagnostics for 1024 candidates and high-pool selections up to pool size 256.

F.3 GENERATION PROCESS

Scenes are generated procedurally from deterministic seeds. Candidate sets are also generated deterministically from target-specific seeds. The generator has no external dependencies beyond NumPy. There are no scraped assets, no personally identifiable data, and no license restrictions on data because data are not collected from users or third-party datasets.

The main artifact is therefore closer to a simulation benchmark than to a static dataset. This has an advantage for auditing: a fresh evaluator can change grid size, pool size, view axes, sampler mixture, and hidden fill scale without collecting new data. The disadvantage is that a synthetic generator may not match the distribution of any real deployment. The paper addresses that by limiting claims and by describing a porting protocol in Appendix J.

F.4 RECOMMENDED REPORTING

A submission using this audit should report at least the following quantities:

- sparse proxy score as a function of candidate pool size;
- true 3D IoU or an equivalent held-out 3D consistency metric;
- hidden-region IoU or hidden-region error;
- selected-candidate mode or a proxy for view-impostor rate;
- candidate-level calibration between proxy score and true utility;
- failure cases for the repaired or proposed selector;
- compute budget, grid resolution, scene count, and random seeds.

Reporting only the best sparse-view proxy score is insufficient because it is exactly the quantity that the audit shows can be misleading.

F.5 KNOWN ARTIFACTS

The candidate generator makes view-impostor candidates by construction. This is not a hidden artifact; it is the failure mode under test. The generator also uses axis-aligned orthographic views, which are easier than posed perspective cameras. This simplification makes the proof and diagnostics transparent but means the benchmark is not a photometric rendering benchmark. The stress suite is compact and should not be used as a high-confidence estimate of small effect sizes. It is a high-pool probe, not a second replicated benchmark.

G APPENDIX G: MASK DERIVATIONS

G.1 PROJECTION OPERATORS

Let an occupancy grid x have shape $m \times m \times m$. For an axis $a \in \{x, y, z\}$, define the projection operator $P_a(x) \in \{0, 1\}^{m \times m}$ by logical OR along that axis. For the x axis,

$$P_x(x)[j, k] = \bigvee_{i=1}^m x[i, j, k].$$

The first-hit index for the same ray is

$$F_x(x)[j, k] = \min\{i : x[i, j, k] = 1\},$$

with a sentinel value when no hit exists. The front-depth map is the first-hit index normalized to $[0, 1]$. The benchmark uses the pair (P_a, F_a) because it captures both silhouette and visible surface depth.

G.2 VISUAL HULL

For a view set $\mathcal{V}$, the visual hull is the intersection of all silhouette extrusions:

$$\mathcal{H}(y; \mathcal{V}) = \bigcap_{a \in \mathcal{V}} \{u \in \mathcal{X} : P_a(y)[\pi_a(u)] = 1\},$$

where π_a maps a voxel to its ray coordinate in view a . The visual hull is an upper bound on target occupancy under silhouette-only constraints. It contains the target but may contain many hidden voxels that are not occupied. With few views, the hull can be much larger than the object.

G.3 VISIBLE MASK

The visible mask is the set of voxels that are first hits from at least one scored view:

$$\text{Visible}(y; \mathcal{V}) = \{u \in y : \exists a \in \mathcal{V}, u_a = F_a(y)[\pi_a(u)]\}.$$

The hidden mask is then $\mathcal{H}(y; \mathcal{V}) \setminus \text{Visible}(y; \mathcal{V})$. This definition avoids penalizing a candidate for voxels outside the sparse-view hull, where the silhouettes already rule out occupancy. It focuses the hidden metric on the ambiguous region where a view impostor can hide errors without changing the proxy.

G.4 NON-INJECTIVITY

For any target with a nonempty hidden mask, construct a candidate x' by copying all visible first-hit voxels and changing any subset of hidden-mask voxels. For every scored ray, the silhouette remains positive exactly when the target silhouette is positive, and the first-hit depth remains unchanged because the copied front voxel is still the first hit. Therefore the proxy score based on silhouette and first-hit depth is identical for all such hidden variants. Their true IoU can vary substantially depending on which hidden voxels are kept or removed. This is the precise sense in which sparse-view projection is not injective.

H APPENDIX H: STATISTICAL INTERPRETATION

H.1 WHY THE REPLICATED RUN CARRIES THE MAIN CLAIM

The replicated benchmark has 72 scenes and evaluates the full pool-size curve up to 64. It is the appropriate evidence for the main statement that increasing candidate pool size improves proxy score while worsening true 3D consistency. The means and standard errors show a large effect: true IoU drops by 0.180 from $N = 1$ to $N = 64$, while proxy score rises by 0.209. The selected-impostor rate increases from 0.208 to 1.000. These are not subtle effects that depend on a single outlier.

H.2 WHY THE HIGH-POOL STRESS IS INTERPRETED DIFFERENTLY

The high-pool stress run is much smaller. It should not be read as a precise estimate of the $N = 256$ population mean. Its role is to test whether proxy saturation, repair improvement, and repair failure can be observed under a more aggressive pool size. It succeeds at that role: the sparse selector saturates the proxy at 1.000, the repair improves true IoU from 0.355 to 0.581, and the failure cases show that the repair remains vulnerable. The paper therefore does not use the compact stress run to replace the replicated result.

H.3 EFFECT-SIZE READING

In the replicated run, the difference between sparse-view and reranked true IoU at $N = 64$ is 0.511. The difference in exploitation gap is 0.533. These are large absolute differences for a bounded metric. The repair’s benefit is not merely that it selects a slightly different candidate; it moves the selected candidate from a nearly pure visible shell toward a more complete occupancy.

The high-pool stress run shows a smaller but still meaningful repair benefit: true IoU improves by 0.226 and the gap drops by 0.226 at $N = 256$. The fact that the selected candidates can still be impostors at $N = 256$ is not a contradiction. It means the repair moves the candidate in the right direction without fully solving the high-pool adversarial case.

H.4 READING UNDEFINED CORRELATIONS

The calibration table reports undefined within-mode correlations for view-impostor candidates. This is not a missing result. In the compact diagnostic set, view-impostor candidates have proxy score 1.000 by construction, so the proxy has no variance within that mode. A correlation coefficient is undefined when one variable is constant. The substantive interpretation is that once candidates reach projective equivalence, the proxy cannot rank them by hidden 3D correctness.

I APPENDIX I: PSEUDOCODE

```

for seed in seeds:
  for sid in scenes:
    y = generate_scene(seed, sid)
    C = sample_candidates(y, max_N)
    for N in pool_sizes:
      P = C[:N]
      F = score_candidates(P, y, axes)
      i_sparse = argmax(F.proxy)
      i_repair = argmax(0.58 * F.proxy
                        + 0.27 * F.consensus
                        + 0.22 * F.volume
                        - 0.12 * F.surface)
      for name, i in selections:
        x = P[i]
        q = F[i].proxy
        u = iou(x, y)
        record(proxy=q, iou3d=u,
              hidden_err=1-hidden_iou(x, y),
              gap=q-u, mode=x.mode)

```

The implementation follows this structure but adds candidate traces, plotting, stress-regime meta-data, and JSON claim audits. The important point is that the repair is evaluated on the same candidate prefix as the sparse-view selector. It does not get a better generator or access to hidden target occupancy.

J APPENDIX J: PORTING THE AUDIT TO NEURAL SCENE SYSTEMS

J.1 PORTING TO NeRF-LIKE MODELS

For a NeRF-like predictor, the target may be represented by held-out multiview images, depth maps, or reconstructed occupancy. Candidate scenes can be different samples from a conditional generative model, different optimization initializations, or different future rollouts. The sparse proxy should match the deployment interface: for example, photometric loss on two observed views. The hidden metric should use held-out views, depth consistency, extracted geometry, or a fused reconstruction. The audit question remains the same: does selecting the best sparse-view candidate improve observed-view loss while worsening held-out 3D or multiview consistency?

J.2 PORTING TO GAUSSIAN SPLATTING

For Gaussian splats, the candidate representation is explicit and renderable. The sparse proxy may be photometric or depth loss under a small camera set. A hidden diagnostic can use novel-view rendering, mesh extraction when available, or occupancy induced by splat opacity. A view-impostor in this setting may be a cloud of splats that explains observed views but places opacity incorrectly in unobserved regions. The repair analog would combine proxy rendering loss with candidate ensemble consensus over opacity, depth uncertainty, and surface regularity.

J.3 PORTING TO AUTONOMOUS-DRIVING OCCUPANCY WORLD MODELS

For occupancy world models, the target may be a future occupancy grid or semantic occupancy volume. Sparse proxy signals may come from camera-visible cells, lidar-visible cells, or task-specific planner reward. Hidden metrics can evaluate occluded occupancy, future collision geometry, and consistency across time. The audit is especially relevant when a planner samples several futures and chooses the one with the highest partial-observation score. A selected future that is plausible in visible cells but wrong in occluded space can create downstream planning risk.

J.4 MINIMUM REAL-SYSTEM PROTOCOL

A real-system version should include at least three candidate pool sizes, at least one sparse proxy matching deployment, at least one held-out geometry or multiview metric, and failure-case visualization. It should report whether the best sparse proxy candidate differs from the best held-out 3D candidate. It should also report whether uncertainty-aware reranking reduces the disagreement without using the held-out metric at selection time.

K APPENDIX K: REVIEWER DECISION TABLE

L APPENDIX L: ADDITIONAL REVIEWER ATTACKS

Attack: The $N = 1$ candidate is not a calibrated baseline. The paper does not require $N = 1$ to be optimal or calibrated. The replicated curve uses $N = 1$ as the smallest pool and shows how selection changes as the same sampler provides more candidates. The effect is visible across the full curve, not only by comparing endpoints.

Attack: Candidate samples are not independent in real models. The proposition uses independence for a clean probability statement. Real candidate samples can be correlated. Correlation changes the exact $1 - (1 - p)^N$ formula but does not remove the audit question. If correlated samples still include hidden-inconsistent projective equivalents, selection can still amplify them. A real-system audit should report effective diversity.

Attack: The repair may be using a prior that happens to match the toy generator. That is possible. The paper therefore calls the repair a baseline and reports mixture, hidden-mass, and ablation stress. The highest-prior stress case shows that the repair can fail. A future learned repair should validate on different scene generators and real data.

Table 8: Common reviewer concerns and the paper’s intended answer.

Concern	Intended answer
Classical geometry already knows sparse views are ambiguous.	Correct. The novelty is the candidate-selection audit and measured pool-size amplification, not the ambiguity itself.
The benchmark is synthetic.	Correct. It is a mechanism benchmark. The paper does not claim real-system failure rates and gives a porting protocol.
The candidate generator includes impostors.	Correct. The audit tests whether selection finds that failure mode as pool size grows.
The repair is hand-designed.	Correct. It is a diagnostic baseline. The ablations and failures show its limits.
More views solve the problem.	More views help when available. The audit targets settings where the deployed score remains sparse or partial.
True 3D IoU is unavailable in deployment.	True IoU is an evaluation diagnostic, not a deployment signal. The selector uses only sparse views and candidate-set geometry.
The high-pool suite is small.	It is a compact stress suite. The replicated 72-scene run carries the main statistical claim.
The paper overclaims.	The claim boundary is explicit: controlled benchmark, scoped mechanism, no universal real-system failure claim.

Attack: Full 3D IoU may not be the task utility. True. Some tasks care about visible rendering or collision-free planning more than volumetric IoU. The audit’s metric should be changed for those tasks. The general requirement is that the held-out metric must measure something the sparse proxy can miss.

Attack: Proxy saturation is an artifact of binary occupancy. Binary occupancy makes saturation easy to inspect, but projective equivalence also appears in continuous opacity and radiance fields under finite views. The exact tolerance and rendering model change, which is why the paper calls for real-system porting rather than claiming direct transfer.

M APPENDIX M: ARTIFACT MANIFEST

The key generated files are:

- `results/full/summary.csv`: replicated 72-scene aggregate metrics up to pool size 64.
- `results/full/metrics.csv`: per-scene replicated metrics.
- `results/expansion/expanded_summary.csv`: compact v3 stress aggregate metrics.
- `results/expansion/expanded_trials.csv`: per-trial v3 stress metrics.
- `results/expansion/candidate_diagnostics.csv`: candidate-level proxy, repair, and true IoU diagnostics.
- `results/expansion/ablation_summary.csv`: repair component ablations.
- `results/expansion/failure_cases.csv`: worst repaired high-pool selections.
- `results/expansion/claims.json`: machine-readable claim audit.
- `paper/final/best-of-n-3d-scene-world-models-v3.pdf`: final repository PDF.

The paper figures are generated into `figures/` and copied into the expansion results directory for self-contained inspection. The build script copies the final v3 PDF to the Desktop under the same v3 filename.

N APPENDIX N: WHY THIS PAPER IS NOT A DUPLICATE WRAPPER

The paper intentionally avoids making the generic claim that selecting the best candidate under an imperfect proxy can overoptimize that proxy. That claim is too broad and would make the paper look like a template applied to a new architecture. The specific contribution here is the geometry of sparse 3D views. The proxy is not merely noisy; it has equivalence classes induced by projection. The failure candidate is not merely high reward; it is a projectively correct visible surface with wrong hidden occupancy. The repair is not a generic verifier; it uses candidate-set geometry, visual-hull mass, and surface structure.

Placed next to papers about language models, decision transformers, diffusion policies, or latent world models, this manuscript should be identifiable from its first page: it is about view impostors, visual hulls, hidden occupancy, and sparse-view 3D scene selection. The abstract, theorem, benchmark, figures, tables, limitations, and appendix all use this geometry-specific vocabulary and evidence. This is the anti-duplication safeguard for the project.

O APPENDIX O: EXPANDED PROOF DETAILS

O.1 A CONCRETE TWO-VOLUME CONSTRUCTION

Consider a single view axis x and a target occupancy y that contains at least one ray with two occupied voxels at depths $i_1 < i_2$. Construct $x^{(a)}$ by keeping both voxels and construct $x^{(b)}$ by keeping only the front voxel at i_1 . The silhouette along that ray is positive for both candidates, and the first-hit depth is i_1 for both candidates. If every other ray is copied from the target, then the single-view proxy gives the same score to $x^{(a)}$ and $x^{(b)}$. Their 3D IoU differs because one candidate contains the hidden voxel and the other does not.

For multiple finite views, the same construction applies to any voxel that is strictly behind the first occupied target voxel for every scored view ray that contains it. Such voxels are exactly the hidden visual-hull voxels used by the benchmark. Changing those voxels does not alter the first-hit surfaces. If the hidden set is nonempty, there are at least $2^{|H|}$ hidden variants with the same visible first-hit pattern. This exponential count is not used as a probability claim; it simply shows that sparse-view equivalence classes can be large.

O.2 WHY CANDIDATE POOLS MATTER

If a selector evaluated one candidate, the sparse proxy could still be wrong, but there would be no selection amplification. With a pool, the selector searches over candidates for high proxy score. If the candidate distribution contains a mode whose hidden geometry is poor but whose proxy score is high, the probability of encountering that mode increases with pool size. The exact increase depends on independence, correlation, and mode probability. The audit does not require the exact formula to hold in real systems. It requires measuring the selected candidate as pool size grows.

O.3 WHY THE REPAIR IS ALLOWED TO USE THE VISUAL HULL

The visual hull is computed from sparse silhouettes. It is therefore available from the same kind of observations used by the proxy. The repair does not know which hidden voxels inside the hull are occupied in the target. It only knows that a candidate whose mass is implausibly small, implausibly large, or concentrated on thin surfaces is suspicious. This is a weak prior, not hidden supervision.

P APPENDIX P: REAL-SYSTEM EXPERIMENT BLUEPRINT

P.1 DATASET SPLIT

A real-system evaluation should separate observed views, selector views, validation views, and hidden evaluation views. Observed views condition the scene predictor. Selector views define the deployed sparse proxy. Validation views tune any repair hyperparameters. Hidden evaluation views

or reconstructed geometry measure whether selected candidates are globally consistent. The hidden split must not be used to choose candidates or tune the final selector.

P.2 CANDIDATE GENERATION

Candidate pools can be generated by stochastic latent sampling, diffusion sampling, dropout, ensemble members, optimization restarts, or explicit multi-hypothesis prediction. The audit should report candidate diversity before selection. If all candidates are nearly identical, a lack of view-impostor amplification may simply mean the generator has low diversity. If candidates are diverse but the proxy still selects poor hidden geometry, the failure is selection-driven.

P.3 METRICS FOR DIFFERENT REPRESENTATIONS

For occupancy systems, use full occupancy IoU, hidden occupancy IoU, semantic IoU if labels exist, and collision-relevant occupancy in occluded regions. For NeRF-like systems, use held-out view photometric error, depth consistency, normal consistency, and geometry extracted through density thresholding. For Gaussian splats, use held-out rendering metrics, depth maps, opacity-derived occupancy, and surface regularity. For point-cloud systems, use Chamfer distance, F-score at multiple thresholds, and occluded-region recall. The metric should be chosen before evaluating candidate selection.

P.4 REQUIRED PLOTS

A real-system paper should include four plots: proxy score versus pool size, held-out 3D consistency versus pool size, exploitation gap versus pool size, and selected-candidate examples at low and high pool sizes. If a repair is proposed, the same plots should compare sparse selection, repaired selection, random selection, and an oracle that selects by the held-out metric. The oracle is not deployable; it shows how much headroom remains.

P.5 STOPPING CRITERIA

The audit should be considered failed if proxy score increases with pool size while held-out 3D consistency decreases, unless the paper explicitly frames that behavior as the discovered failure. A repair should be considered insufficient if it improves the mean but leaves high-severity hidden failures unreported. The benchmark should report both averages and worst cases.

Q APPENDIX Q: FAILURE TAXONOMY

Thin-shell impostor. The candidate copies only front surfaces and leaves the interior empty or sparse. It usually has high visible occupancy fraction and high surface fraction. The current repair targets this case through surface penalty and volume plausibility.

False-hidden-mass impostor. The candidate copies front surfaces but fills hidden visual-hull regions with wrong mass. It may evade a simple volume prior if the occupied count is plausible. Candidate consensus or additional views are needed to detect it.

Consensus impostor. If many candidates share the same hidden error, ensemble consensus can reinforce the wrong geometry. This is a serious limitation for any consensus-based repair. A real-system repair should estimate epistemic uncertainty and compare against priors from training data, not only within-pool agreement.

View-allocation failure. The sparse view set may miss exactly the region where candidate futures differ. In this case reranking may be less effective than active sensing. The system should request a new view that intersects the disagreement region.

Proxy-overconfident faithful drift. Not every failure is an explicit view impostor. A faithful-mode candidate can drift in a way that preserves the sparse projections while changing hidden geometry. The audit still catches this through hidden error and exploitation gap, even if the generator-mode label is not “view impostor.”

R APPENDIX R: FIFTY-ROUND REVIEWER CHECKLIST CONDENSED

The following checklist summarizes the self-attack process used to harden the paper:

1. Is the title distinct from generic candidate-selection papers? Yes.
2. Does the abstract name the 3D mechanism? Yes.
3. Are claims scoped to synthetic evidence? Yes.
4. Is classical visual hull work acknowledged? Yes.
5. Is proxy overoptimization prior work acknowledged? Yes.
6. Does the paper avoid claiming a new renderer? Yes.
7. Does the selector avoid hidden labels? Yes.
8. Does the repair avoid hidden labels? Yes.
9. Are pool-size curves reported? Yes.
10. Are hidden-region metrics reported? Yes.
11. Are selected-impostor rates reported? Yes.
12. Are standard errors reported in CSV summaries? Yes.
13. Is the high-pool stress suite labeled compact? Yes.
14. Does the paper avoid using compact stress as the main claim? Yes.
15. Are repair failures reported? Yes.
16. Are ablations included? Yes.
17. Is candidate-level calibration included? Yes.
18. Is random selection included in the stress suite? Yes.
19. Are axis-count effects included? Yes.
20. Are sampler-mixture effects included? Yes.
21. Are hidden-mass effects included? Yes.
22. Is compute described? Yes.
23. Are deterministic seeds used? Yes.
24. Is the final PDF built by script? Yes.
25. Is the Desktop artifact named v3? Yes.
26. Is the source folder the source of truth? Yes.
27. Are stale version-two names excluded from the manuscript? Yes.
28. Is there a machine-readable claim audit? Yes.
29. Does the paper state what it does not claim? Yes.
30. Does it give a real-system porting path? Yes.
31. Does it include enough detail to reproduce the toy benchmark? Yes.
32. Does it explain why impostor candidates are included? Yes.
33. Does it explain why more views help but do not erase the audit? Yes.
34. Does it separate visible and hidden signals conceptually? Yes.
35. Does it include reviewer attack responses? Yes.
36. Does it include a failure taxonomy? Yes.
37. Does it avoid relying on a single figure? Yes.
38. Does it include both means and qualitative examples? Yes.
39. Does it identify repair headroom? Yes.
40. Does it avoid universal deployment claims? Yes.
41. Does it avoid duplicate wrapper language? Yes.

42. Does it use geometry-specific notation? Yes.
43. Does it explain visual hull construction? Yes.
44. Does it discuss correlation limitations? Yes.
45. Does it say why undefined correlations are meaningful? Yes.
46. Does it document generated artifacts? Yes.
47. Does it mention CPU/RAM constraints honestly? Yes.
48. Does it provide exact commands? Yes.
49. Does it provide a final acceptance audit? Yes.
50. Does the paper remain falsifiable? Yes.

S APPENDIX S: ADDITIONAL DISCUSSION OF COMPUTE CONSTRAINTS

The v3 pass attempted larger high-pool stress configurations, but pure Python scoring at $N = 256$ over many scenes was too slow on the available machine while other background experiments were active. Rather than hide this, the final artifact makes the engineering choice explicit. The replicated 72-scene benchmark remains the main quantitative result. The high-pool suite is compact but covers more adversarial axes: pool size 256, axis count, mixture shift, hidden fill, ablation, calibration, and failures. This is the correct submission stance because it maximizes rigor under the actual compute budget without pretending that a small stress suite has the confidence of a large benchmark.

Future work should implement vectorized scoring or use sparse voxel data structures so that high-pool replication can be expanded without increasing RAM. The most expensive operations are repeated candidate scoring and consensus recomputation across prefixes. A production benchmark can precompute view projections, cache per-candidate proxy features, and update consensus incrementally. Those engineering optimizations are orthogonal to the audit claim, but they would allow a larger $N = 256$ or $N = 512$ study.

T APPENDIX T: CLOSEST PRIOR-WORK SEPARATIONS

Sparse-view reconstruction papers. Sparse-view NeRF and occupancy papers usually ask how to reconstruct a scene from limited observations. This paper asks a different question: after a model has produced several possible scenes, what happens when an inference-time selector chooses among them with a sparse proxy? A strong sparse-view reconstruction method can still be placed inside a candidate-selection interface, and the audit would then test whether its sampled candidates are ranked safely.

Uncertainty and ensemble papers. Uncertainty methods often use ensembles to estimate epistemic uncertainty. The repair here borrows the simplest possible ensemble-consensus idea but does not claim a new uncertainty estimator. The contribution is to show that a geometric uncertainty signal is relevant to the diagnosed failure, and to expose where that signal fails at high pool size.

Verifier-guided generation. Verifier-guided generation papers select among candidates with a learned or hand-designed verifier. The analogy is useful, but the verifier failure here has a specific 3D structure. A verifier can be wrong because it is noisy or miscalibrated. A sparse-view 3D proxy can be wrong because many hidden volumes are exactly equivalent under the observed projections. That equivalence is the reason a candidate can be proxy-perfect and still geometrically poor.

Classical visual hull work. Classical visual hull work explains why sparse silhouettes do not determine shape. This paper assumes that fact and asks how it interacts with learned candidate pools. The empirical artifact is therefore not a reproof of visual hull ambiguity; it is a stress test for modern scene-world interfaces that sample, score, and select.

Planning with learned world models. Planning papers often evaluate whether model rollouts improve downstream task reward. A planner may not care about every hidden voxel. However, if hidden geometry affects collision, reachability, or future observations, then a sparse-view-selected

scene can be dangerous even when the immediate proxy looks good. The audit can be adapted by replacing true IoU with a task-specific hidden utility.

Why the paper should survive side-by-side comparison. The manuscript is anchored by view impostors, visual hulls, hidden occupancy, first-hit projections, and selected-candidate 3D IoU. These are not cosmetic renamings of another architecture’s candidate-selection story. The theorem, benchmark, repair, stress tests, and failure taxonomy all depend on 3D projective ambiguity. If a reviewer places this paper beside another candidate-pool paper, the mechanism, metrics, figures, and limitations should make the identity of this work immediately clear.

U APPENDIX U: DEPLOYMENT CHECKLIST FOR FUTURE USERS

A team applying this audit to a real scene-world model should answer the following questions before claiming that larger candidate pools improve the system.

What is the deployed proxy? The proxy must be the score that actually chooses candidates. If a system selects by two camera views and reports success on ten held-out views, then the audit must preserve that difference. Evaluating every candidate with the held-out views would hide the selection failure.

What hidden quantity can the proxy miss? For a robot, this may be occluded collision geometry. For a renderer, it may be novel-view consistency. For an occupancy predictor, it may be cells outside the sensor frustum. The hidden quantity should be named before experiments are run.

How diverse are the candidates? Candidate-pool amplification requires meaningful variation. If the sampler produces near-duplicates, the audit may show little effect even though the proxy is incomplete. Diversity should be measured before selection and among the top-proxy candidates after selection.

Does the proxy improve while hidden consistency worsens? This is the decisive pattern. If both proxy and hidden consistency improve, larger pools are helping. If proxy improves and hidden consistency falls, the system is overselecting for the sparse view interface.

Does a repair reduce mean failure and worst-case failure? A repair that improves only the mean can still leave dangerous hidden cases. The audit should report worst selected examples, not only averages.

Would another view resolve the disagreement? If candidate disagreement is concentrated in hidden visual-hull regions, active view acquisition may be better than reranking. A safe scene-world interface can return “need more evidence” rather than forcing a single selected future.

Are the claims stated at the right level? If the experiment is synthetic, claim a mechanism. If the experiment uses a trained model on a real dataset, claim an empirical failure rate for that setting only. If the repair is heuristic, report it as a baseline. This discipline is what keeps the audit useful rather than becoming another overbroad candidate-selection story.

REFERENCES

- Mildenhall, B., Srinivasan, P., Tancik, M., Barron, J., Ramamoorthi, R., and Ng, R. NeRF: Representing scenes as neural radiance fields for view synthesis. ECCV, 2020.
- Sitzmann, V., Zollhoefer, M., and Wetzstein, G. Scene representation networks: Continuous 3D-structure-aware neural scene representations. NeurIPS, 2019.
- Yu, A., Ye, V., Tancik, M., and Kanazawa, A. pixelNeRF: Neural radiance fields from one or few images. CVPR, 2021.
- Wang, Q., Wang, Z., Genova, K., Srinivasan, P., Zhou, H., Barron, J., Martin-Brualla, R., Snavely, N., and Funkhouser, T. IBRNet: Learning multi-view image-based rendering. CVPR, 2021.
- Niemeyer, M., Barron, J., Mildenhall, B., Sajjadi, M., Geiger, A., and Radwan, N. RegNeRF: Regularizing neural radiance fields for view synthesis from sparse inputs. CVPR, 2022.

- Wang, G., Chen, Z., Loy, C., and Liu, Z. SparseNeRF: Distilling depth ranking for few-shot novel view synthesis. ICCV, 2023.
- Kerbl, B., Kopanas, G., Leimkuehler, T., and Drettakis, G. 3D Gaussian Splatting for real-time radiance field rendering. SIGGRAPH, 2023.
- Guedon, A. and Lepetit, V. SuGaR: Surface-aligned Gaussian Splatting for efficient 3D mesh reconstruction. CVPR, 2024.
- Huang, B., Yu, Z., Chen, A., Geiger, A., and Gao, S. 2D Gaussian Splatting for geometrically accurate radiance fields. SIGGRAPH, 2024.
- Mescheder, L., Oechsle, M., Niemeyer, M., Nowozin, S., and Geiger, M. Occupancy Networks: Learning 3D reconstruction in function space. CVPR, 2019.
- Peng, S., Niemeyer, M., Mescheder, L., Pollefeys, M., and Geiger, A. Convolutional Occupancy Networks. ECCV, 2020.
- Cao, A. and de Charette, R. MonoScene: Monocular 3D semantic scene completion. CVPR, 2022.
- Li, Y., Yu, Z., Choy, C., Xiao, C., Alvarez, J., Fidler, S., Feng, C., and Anandkumar, A. VoxFormer: Sparse voxel transformer for camera-based 3D semantic scene completion. CVPR, 2023.
- Zheng, W. et al. OccWorld: Learning a 3D occupancy world model for autonomous driving. arXiv, 2024.
- Guzman-Rivera, A., Batra, D., and Kohli, P. Multiple choice learning: Learning to produce multiple structured outputs. NeurIPS, 2012.
- Bhattacharyya, A., Fritz, M., and Schiele, B. Best-of-many samples objective for generative models. CVPR, 2018.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. arXiv, 2021.
- Lightman, H. et al. Let’s verify step by step. ICLR, 2023.
- Laurentini, A. The visual hull concept for silhouette-based image understanding. IEEE TPAMI, 1994.
- Kutulakos, K. and Seitz, S. A theory of shape by space carving. IJCV, 2000.